Midterm Review

COMP2711 Discrete Mathematical Tools

Mingxun Zhou (mingxunz@ust.hk)

Midterm coverage and difficulty

Coverage

- All midterm questions are covered by our lectures.
- This short review covers most of the material, but is not exhaustive.

Difficulty

- Basic concepts and straightforward calculations.
- Some questions require more thought and calculation.
- A small number of questions are more challenging.

Exam strategy

1. Understand the homework problems and their solutions, even if you did not solve them yourself.
2. Scan the whole exam and identify questions you can solve readily.
3. Secure the basic points first.
4. Time will be tight. Do not spend too long on one question. If you get stuck, move on and return later.

Propositional logic

Formula: propositions combined using logical connectives.

$$\neg p, \quad p \wedge q, \quad p \vee q, \quad p \oplus q, \quad p \to q, \quad p \leftrightarrow q.$$

Tautology True for every truth assignment.

Contradiction False for every truth assignment.

$A \equiv B$: the same truth value for every assignment.

Equivalently, $A \leftrightarrow B$ is a tautology.

Satisfiable: true for at least one truth assignment.

Logical equivalence laws

Truth constants	$p \wedge T \equiv p, \quad p \vee F \equiv p$
	$p \wedge F \equiv F, \quad p \vee T \equiv T$
Repetition	$p \wedge p \equiv p, \quad p \vee p \equiv p$
Opposites	$p \wedge \neg p \equiv F, \quad p \vee \neg p \equiv T$
Double negation	$\neg\neg p \equiv p$
De Morgan	Negate each part and flip AND $\leftrightarrow$ OR.
	$\neg(p \wedge q) \equiv \neg p \vee \neg q$
	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Distribution	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

Implication and biconditional

Implication

$$p \to q \equiv \neg p \vee q.$$

When the implication is false

$$\neg(p \to q) \equiv \underbrace{p \wedge \neg q}_{p=T \text{ and } q=F}.$$

Contraposition

$$p \to q \equiv \neg q \to \neg p.$$

Biconditional

$$\begin{aligned} p \leftrightarrow q &\equiv (p \to q) \wedge (q \to p) \\ &\equiv (p \wedge q) \vee (\neg p \wedge \neg q). \end{aligned}$$

The converse $q \to p$ is not generally equivalent to $p \to q$.

Predicate logic

Domain: the objects over which variables range.

$$\forall x\, P(x):\ \text{every } x \text{ satisfies } P, \qquad \exists x\, P(x):\ \text{some } x \text{ satisfies } P.$$

A variable is **bound** within its quantifier's scope; otherwise it is **free**.

Negating quantified formulas

$$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x),$$
$$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x).$$

Reordering quantifiers

1. **Same type:** either order is equivalent.

$$\forall x \forall y\, R(x, y) \equiv \forall y \forall x\, R(x, y),$$
$$\exists x \exists y\, R(x, y) \equiv \exists y \exists x\, R(x, y).$$

2. **Mixed type:** only one direction holds in general.

$$\exists y\, \forall x\, R(x, y) \implies \forall x\, \exists y\, R(x, y).$$

Quantified formulas

Important: express natural-language statements using logical formulas.

Restricted domains

$$\text{Every } A \text{ is } B: \quad \forall x(A(x) \to B(x)).$$

$$\text{Some } A \text{ is } B: \quad \exists x(A(x) \wedge B(x)). \quad \text{An object is both } A \text{ and } B.$$

Uniqueness

$$\exists x(P(x) \wedge \forall y(P(y) \to y = x)).$$

Inference and proof methods

No need to memorize the names. These are natural rules of reasoning.

Modus ponens	$P,\ P \to Q \ \Rightarrow Q$
Modus tollens	$\neg Q,\ P \to Q \ \Rightarrow \neg P$
Resolution	$P \vee R,\ \neg P \vee Q \ \Rightarrow R \vee Q$
Direct proof	Assume P and derive Q.
Contraposition	To prove $P \to Q$, equivalently prove $\neg Q \to \neg P$.
Contradiction	Negate the target and derive a contradiction.
Cases	Cover all possibilities and prove the target in each.
Existence and uniqueness	Show at least one, then show at most one.
Equivalence	Prove both directions (or a cycle for several claims).

Sets

$$\text{Set inclusion:} \quad A \subseteq B \iff \forall x(x \in A \to x \in B),$$
$$\text{Proper inclusion:} \quad A \subsetneq B \iff A \subseteq B \text{ and } A \neq B.$$

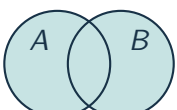
$$\text{Cartesian product:} \quad A \times B = \{(a, b) : a \in A,\ b \in B\},$$
$$\text{Power set:} \quad \mathcal{P}(A) = \{B : B \subseteq A\}.$$

For finite sets:

$$|\varnothing| = 0, \quad |A \times B| = |A||B|, \quad |\mathcal{P}(A)| = 2^{|A|}.$$

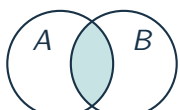
Set operations

Union: $A \cup B$



$x \in A$ or $x \in B$

Intersection: $A \cap B$



$x \in A$ and $x \in B$

Difference: $A \setminus B$



$x \in A$ and $x \notin B$

Set identities

Complements are relative to a fixed universe U.

$$\left.\begin{aligned}
\overline{A \cup B} &= \overline{A} \cap \overline{B} \\
\overline{A \cap B} &= \overline{A} \cup \overline{B} \\
A \cap (B \cup C) &= (A \cap B) \cup (A \cap C) \\
A \cup (B \cap C) &= (A \cup B) \cap (A \cup C) \\
A \cup \varnothing &= A, \qquad A \cap U = A \\
A \cup \overline{A} &= U, \qquad A \cap \overline{A} = \varnothing
\end{aligned}\right\}$$

Logical equivalence laws translated into set language

$\vee \leftrightarrow \cup, \quad \wedge \leftrightarrow \cap$

$\neg \leftrightarrow$ complement

Proving set equality

$$A = B \iff (A \subseteq B \text{ and } B \subseteq A).$$

Equivalently, prove $x \in A \iff x \in B$ for an arbitrary x.

Functions

A function $f : A \to B$ assigns each $a \in A$ exactly one value in B.

$$\text{Domain: } A, \quad \text{codomain: } B, \quad \text{range: } \{f(a) : a \in A\}.$$

$$\begin{aligned} \text{Injective: } & \forall x, y \in A,\ f(x) = f(y) \Rightarrow x = y. \\ & \text{Equivalently, } x \neq y \Rightarrow f(x) \neq f(y). \\ \text{Surjective: } & \forall b \in B\ \exists a \in A :\ f(a) = b. \\ \text{Bijective: } & \text{injective and surjective.} \end{aligned}$$

Composition: $(g \circ f)(x) = g(f(x))$; associative, generally not commutative.

Inverse: $f^{-1} : B \to A$ exists exactly when f is bijective.

$$f^{-1} \circ f = \mathrm{id}_A, \qquad f \circ f^{-1} = \mathrm{id}_B .$$

Cardinality

$\lvert A\rvert \le \lvert B\rvert$	An injection $A \to B$ exists.
$\lvert A\rvert = \lvert B\rvert$	A bijection $A \to B$ exists.
$\lvert A\rvert < \lvert B\rvert$	$\lvert A\rvert \le \lvert B\rvert$, but no bijection exists.
Schröder–Bernstein	$\lvert A\rvert \le \lvert B\rvert$ and $\lvert B\rvert \le \lvert A\rvert$ imply $\lvert A\rvert = \lvert B\rvert$.
Countable	Finite or in bijection with $\mathbb{N}$.
Uncountable	Cardinality strictly greater than $\lvert\mathbb{N}\rvert$.

$$|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}| = \aleph_0, \qquad |\mathbb{R}| = |\mathcal{P}(\mathbb{N})| > \aleph_0.$$

If A, B are countable, so are $A \cup B$, $A \cap B$, $A \setminus B$, and $A \times B$.

Cantor's theorem: $|S| < |\mathcal{P}(S)|$ for every set S.

Basic counting

Sum rule	Disjoint cases: $N = N_1 + \cdots + N_k$.
Product rule	Successive choices: $N = n_1 \cdots n_k$ if every partial choice has n_i options at step i.
Complement	$\lvert\text{valid}\rvert = \lvert\text{all}\rvert - \lvert\text{invalid}\rvert$.
Division rule	N/k outcomes if N descriptions count each outcome exactly k times.

Generalized pigeonhole principle

$$N \text{ objects in } k \geq 1 \text{ boxes} \implies \text{some box has at least } \lceil N/k \rceil .$$

To force at least $r \geq 1$ objects in one box: $k(r-1)+1$ objects suffice.

Permutations and combinations

Choose $r \geq 0$ objects from $n \geq 1$ distinct types.

	Repetition allowed	**No repetition ($r \leq n$)**
Order matters	n^r 4-digit PINs; digits may repeat.	$P(n,r) = \dfrac{n!}{(n-r)!}$ Arrange 3 of 10 people in a photograph.
Order ignored	$\binom{r+n-1}{n-1}$ Stars and bars	$\binom{n}{r} = \dfrac{n!}{r!(n-r)!}$ Choose a 3-person team from 10 people.

Circular arrangements: $(n-1)!$ for n distinct objects; rotations identified, reflections distinct.

Stars and bars

For integers $x_1, \ldots, x_n$, where $n \geq 1$ and $r \geq 0$:

$$x_1 + \cdots + x_n = r.$$

Nonnegative solutions: r stars and $n - 1$ bars.

$$\#\{(x_1, \ldots, x_n) : x_i \geq 0\} = \binom{r + n - 1}{n - 1}.$$

Positive solutions: subtract 1 from each variable first.

$$\#\{(x_1, \ldots, x_n) : x_i \geq 1\} = \binom{r - 1}{n - 1} \quad (r \geq n).$$

There are no positive solutions when $r < n$.

Binomial coefficients

$$(ax + by)^n = \sum_{k=0}^{n} \binom{n}{k} (ax)^{n-k} (by)^k.$$

Complement $\binom{n}{r} = \binom{n}{n-r}$

Pascal $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$

Hockey-stick $\sum_{k=r}^{n} \binom{k}{r} = \binom{n+1}{r+1}$

Vandermonde $\sum_k \binom{a}{k}\binom{b}{r-k} = \binom{a+b}{r}$

Integer parameters; $n \geq 0$ ($n \geq 1$ for Pascal), $0 \leq r \leq n$ for complement and hockey-stick. In Vandermonde, $a, b, r \geq 0$; out-of-range binomial coefficients are 0.

Inclusion–exclusion

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

$$\left|\bigcup_{i=1}^{m} A_i\right| = \sum_{k=1}^{m} (-1)^{k-1} \sum_{1 \le i_1 < \cdots < i_k \le m} |A_{i_1} \cap \cdots \cap A_{i_k}|.$$

Count OR conditions using counts of AND conditions.

To avoid all bad conditions: subtract their union from all outcomes.

Surjections (m inputs, n outputs) $\displaystyle\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m$

Derangements of n objects $\displaystyle n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$

Here $m, n \ge 1$.

Probability

Sample space S: all outcomes. **Event E:** a subset of S.

$$\Pr[E] \geq 0, \qquad \Pr[S] = 1.$$

For pairwise disjoint events, the probability of their union is the sum.

$$\Pr[\overline{E}] = 1 - \Pr[E].$$

$$\Pr[E \cup F] = \Pr[E] + \Pr[F] - \Pr[E \cap F].$$

Classical model: finite, equally likely outcomes.

$$\Pr[E] = \frac{|E|}{|S|}.$$

Reducing the sample space preserves uniformity when every retained outcome represents the same number of original outcomes.

Conditional probability and independence

$$\Pr[E \mid F] = \frac{\Pr[E \cap F]}{\Pr[F]} \qquad (\Pr[F] > 0).$$

Multiplication rule

$$\Pr[E \cap F] = \Pr[E \mid F] \Pr[F].$$

Independence: definition

$$E, F \text{ independent} \iff \Pr[E \cap F] = \Pr[E] \Pr[F].$$

When $\Pr[F] > 0$, equivalently $\Pr[E \mid F] = \Pr[E]$.

If E, F are independent, so are $E, \overline{F}$; $\overline{E}, F$; and $\overline{E}, \overline{F}$.

Conditional probability specifies neither causality nor time order.

Total probability and Bayes' theorem

Let $F_1, \ldots, F_m$ partition S, with $\Pr[F_i] > 0$.

Total probability

$$\Pr[E] = \sum_{i=1}^{m} \Pr[E \mid F_i] \Pr[F_i].$$

Bayes' theorem ($\Pr[E] > 0$)

$$\Pr[F_j \mid E] = \frac{\Pr[E \mid F_j] \Pr[F_j]}{\sum_{i=1}^{m} \Pr[E \mid F_i] \Pr[F_i]}.$$

Prior	$\Pr[F_j]$
Likelihood of the evidence	$\Pr[E \mid F_j]$
Posterior	$\Pr[F_j \mid E]$

Random variables and distributions

Random variable: a function $X : S \to \mathbb{R}$.

For a discrete sample space:

$$\Pr[X = t] = \sum_{\omega : X(\omega) = t} \Pr[\{\omega\}].$$

A discrete distribution lists each possible value and its probability.

$$\Pr[X = t] \geq 0, \qquad \sum_{t} \Pr[X = t] = 1.$$

Independent discrete random variables

$$\Pr[X = x,\ Y = y] = \Pr[X = x] \Pr[Y = y] \quad \text{for all } x, y.$$

Discrete distributions

Distribution	Probability	Values
Uniform on n values	$\Pr[X = t] = 1/n$	Each allowed t
Bernoulli(p)	$\Pr[X = 1] = p,\ \Pr[X = 0] = 1 - p$	$0, 1$
Geometric(p)	$\Pr[X = k] = (1 - p)^{k-1} p$	$k = 1, 2, \ldots$

Geometric: trial number of the first Yes in independent trials ($p > 0$).

Binomial: Yes count in n independent trials with Yes probability p.

$$X \sim \mathrm{Bin}(n, p), \qquad \Pr[X = k] = \binom{n}{k} p^k (1 - p)^{n-k}, \quad 0 \le k \le n.$$

Expectation

$$\mathbb{E}[X] = \sum_t t \Pr[X = t], \qquad \mathbb{E}[g(X)] = \sum_t g(t) \Pr[X = t].$$

Linearity: no independence required

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

Indicators: $I_A = 1$ when A occurs, and 0 otherwise.

$$\mathbb{E}[I_A] = \Pr[A], \qquad \mathbb{E}\left[\sum_{i=1}^{n} I_{A_i}\right] = \sum_{i=1}^{n} \Pr[A_i].$$

$$X \sim \mathrm{Bin}(n, p) \quad \Longrightarrow \quad \mathbb{E}[X] = np.$$

In general, $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$ and $\mathbb{E}[XY] \neq \mathbb{E}[X]\mathbb{E}[Y]$.